Code No: 134BU **R16**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B.Tech II Year II Semester Examinations, March - 2022
OPERATING SYSTEMS
(Common to CSE, IT)

Time: 3 Hours **Max. Marks: 75**

Answer any five questions
All questions carry equal marks
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1.a) Explain the Operating System services.
b) Discuss the Protection and Security. [8+7]

2. List and explain the various types of system calls. [15]

3.

		PID	AT	BT
		CPU	IO	CPU
P1	0	1	2	2
P2	1	2	4	5
P3	2	3	6	8

What are the completion times of processes P1, P2 and P3 using SRTF (Shortest Remaining Time First)? [15]

4.a) Draw and explain the process state diagram with two suspend states.
b) Explain the Classic Problems of Synchronization. [8+7]

5.a) In a paged memory, the page hit ratio is 0.35. The time required to access a page in secondary memory is equal to 100nS. The time required to access a page in physical memory is 10nS. What is the average time required to access a page?
b) Discuss the hardware support required to support demand paging. [8+7]

6. Explain about FIFO, LRU page replacement algorithms with example. [15]

7.a) Discuss the Directory and Disk Structure.
b) Give a brief note on File System Structure. [8+7]

8.a) Define deadlock. Explain the necessary conditions for a deadlock to occur.
b) Is it possible to apply the Resource allocation graph algorithm to avoid deadlock? Explain. [7+8]

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